

JUnit Testing Exercises

Exercise 1: Setting Up JUnit

Scenario:

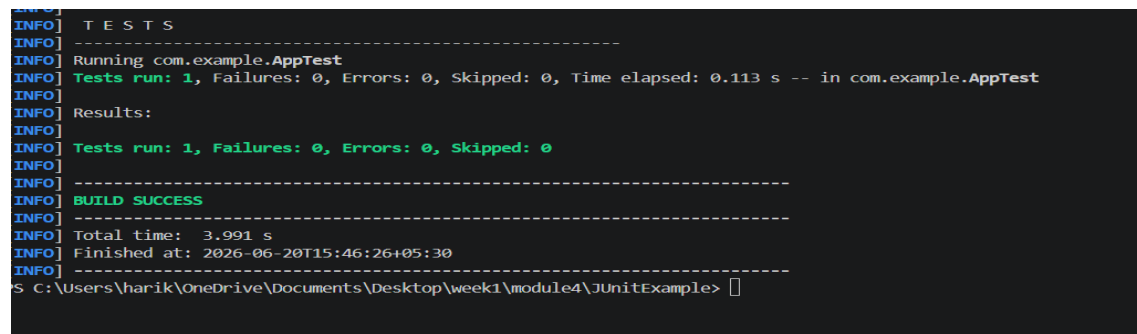
You need to set up JUnit in your Java project to start writing unit tests.

Steps:

1. Create a new Java project in your IDE (e.g., IntelliJ IDEA, Eclipse).
2. Add JUnit dependency to your project. If you are using Maven, add the following to your pom.xml:

```
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>4.13.2</version>
  <scope>test</scope>
</dependency>
```

3. Create a new test class in your project.



```
INFO] TESTS
INFO] -----
INFO] Running com.example.AppTest
INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.113 s -- in com.example.AppTest
INFO] Results:
INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
INFO] -----
INFO] BUILD SUCCESS
INFO] -----
INFO] Total time: 3.991 s
INFO] Finished at: 2026-06-20T15:46:26+05:30
INFO] -----
S C:\Users\harik\OneDrive\Documents\Desktop\week1\module4\JUnitExample>
```

Exercise 2: Writing Basic JUnit Tests

Scenario:

You need to write basic JUnit tests for a simple Java class.

Steps:

1. Create a new Java class with some methods to test.
2. Write JUnit tests for these methods.

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.116 s -- in com.example.AppTest
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in com.example.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.519 s
[INFO] Finished at: 2026-06-20T21:57:09+05:30
[INFO] -----
```

Exercise 3: Assertions in JUnit

Scenario:

You need to use different assertions in JUnit to validate your test results.

Steps:

1. Write tests using various JUnit assertions.

Solution Code:

```
public class AssertionsTest {
    @Test public void
    testAssertions() {
        // Assert equals
        assertEquals(5, 2 + 3);

        // Assert true
        assertTrue(5 > 3);

        // Assert false
        assertFalse(5 < 3);

        // Assert null
        assertNull(null);
    }
}
```

```

    // Assert not null
    assertNotNull(new Object());
}
}

```

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.138 s -- in com.example.AppTest
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.004 s -- in com.example.AssertionsTest
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.002 s -- in com.example.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]

```

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

Scenario:

You need to organize your tests using the Arrange-Act-Assert (AAA) pattern and use setup and teardown methods.

Steps:

1. Write tests using the AAA pattern.
2. Use `@Before` and `@After` annotations for setup and teardown methods.

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.134 s -- in com.example.AppTest
[INFO] Running com.example.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.AssertionsTest
[INFO] Running com.example.CalculatorAAATest
[INFO] Setup completed
[INFO] Teardown completed
[INFO] Setup completed
[INFO] Teardown completed
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.011 s -- in com.example.CalculatorAAATest
[INFO] Running com.example.CalculatorTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in com.example.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.286 s
[INFO] Finished at: 2020-06-20T22:31:25+05:30
[INFO]

```